

UK Space-Based Solar Power Cost-Condition Assessment — v2.0

This is a conditional scenario result, not a commercial forecast, quotation, official UK target or proof of economic viability.

Scope and answer

The reference scenario gives a conditional delivered-grid DCF LCOE of **£86.03/MWh** in 2024 real GBP. Working backwards from 2 GW AC at the grid boundary, the five-stage chain computes an end-to-end efficiency of **14.98%**, orbital hardware mass of **10.00 million kg**, and **118** equivalent launches.

Launch pricing mode: per kg, to staging orbit only. Staging orbit: LEO staging orbit (service boundary; altitude architecture-dependent). Operational orbit: High Earth orbit (HEO; final orbit architecture-dependent). The transfer proxy must cover transfer vehicle, propellant, refuelling missions, operations and payload-performance penalty.

Migration from v1.x to v2.0

v1.x multiplied a 1.5 kg/kW value that should have been delivered-power-normalised by 'delivered power / efficiency', dividing by efficiency a second time. For 2 GW at 20% efficiency, the old expression gives **15,000,000 kg (15,000 tonnes)**; the corrected expression gives **3,000,000 kg (3,000 tonnes)**. With all other v2 comparison inputs identical, the erroneous mass boundary gives **£116.45/MWh**, versus **£43.45/MWh** on the corrected boundary. Historical v1.x thresholds are therefore not directly comparable with v2.0.

The DESNZ/Frazer-Nash report defines architecture specific power as ground-delivered power per orbital mass. The reciprocal of 0.67 kW-delivered/kg is **1.4925 kg/kW-delivered** (Table 1, PDF pp.11-12; reference-design selection p.13). Values remain architecture-specific and may include manufacturer claims that were not independently verified.

Stage-resolved energy chain

Stage	Rated power
Incident solar	13.35 GW
Space DC bus	4.67 GW
Emitted RF	3.27 GW
RF incident on rectenna	3.21 GW
Rectenna DC output	2.08 GW
Grid-delivered AC	2.00 GW

Computed end-to-end efficiency: **14.9822%**.

Discounted-cash-flow boundary

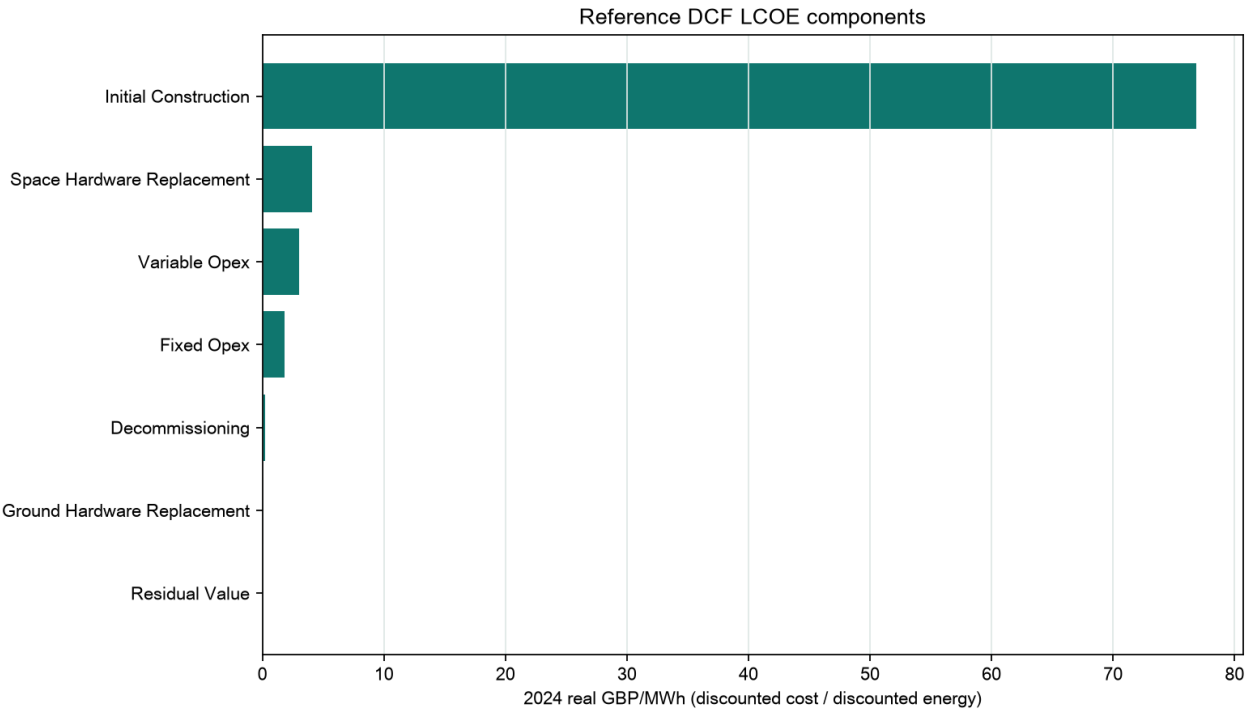
The headline uses $LCOE = \Sigma(C_t / (1+r)^t) / \Sigma(E_t / (1+r)^t)$. The valuation base is start of construction at $t=0$; four equal construction shares are spent before commissioning, and the first operating-year cash flow is at $t=5$. The operating life is 30 years. First-year energy is 15.77 TWh and lifetime-average annual energy is 14.68 TWh. Discounted lifecycle cost is £13.10bn and discounted energy is 152.32 million MWh. The simple CRF reconciliation is £71.02/MWh and is secondary only.

Reference cost structure

CAPEX component	2024 real GBP
Launch To Staging Orbit	£5.00bn
In Orbit Assembly And Deployment	£2.00bn
Space Generation Hardware	£1.87bn
Orbit Transfer To Operational Orbit	£1.00bn
Wireless Power Transmitter	£0.33bn
Rectenna	£0.30bn
Grid Connection	£0.20bn
Programme contingency	£2.14bn
Initial CAPEX	£12.84bn

Lifecycle component	Discounted PV	LCOE contribution
Initial Construction	£11.71bn	£76.86/MWh
Space Hardware Replacement	£0.62bn	£4.08/MWh
Variable Opex	£0.46bn	£3.00/MWh
Fixed Opex	£0.27bn	£1.80/MWh
Decommissioning	£0.03bn	£0.20/MWh
Ground Hardware Replacement	£0.02bn	£0.10/MWh
Residual Value	£-0.00bn	£-0.00/MWh

Reference lifecycle LCOE components



Conditional sensitivities and thresholds

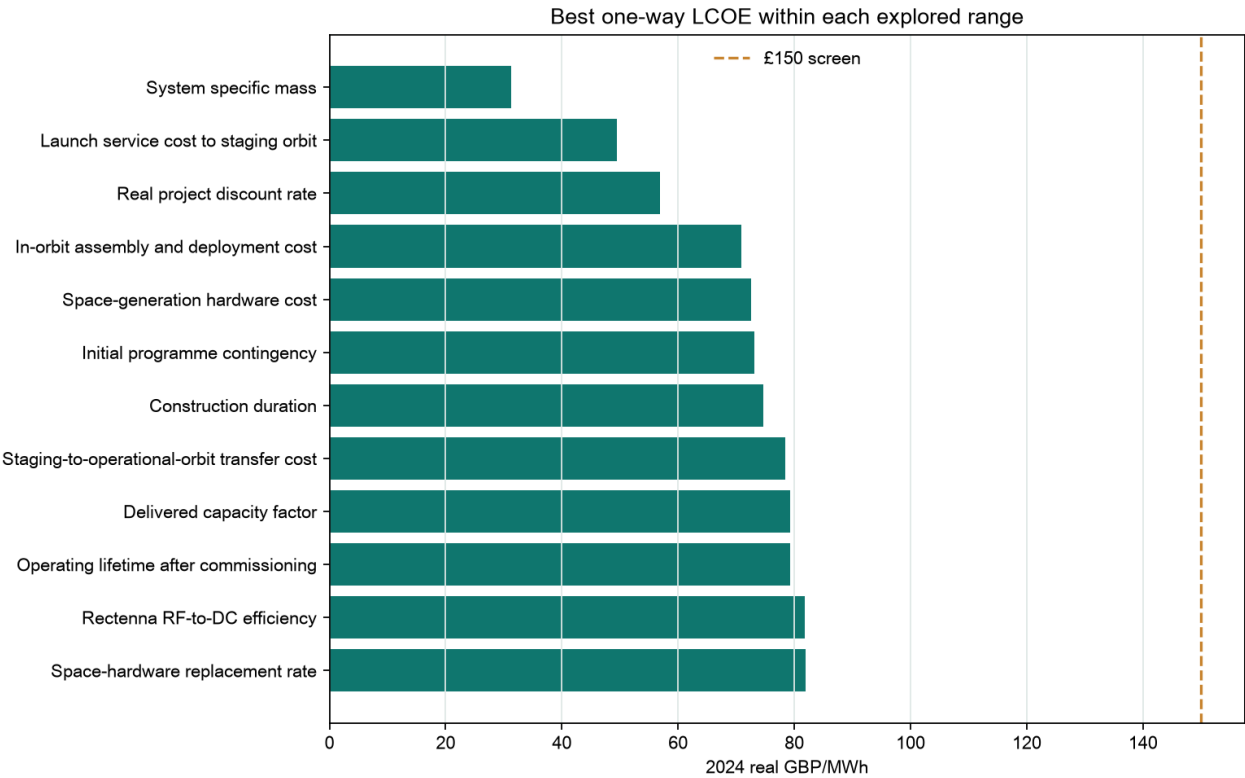
Parameter	LCOE at favourable bound	Reduction
System specific mass	£31.3/MWh	£54.7/MWh
Launch service cost to staging orbit	£49.5/MWh	£36.5/MWh
Real project discount rate	£56.9/MWh	£29.1/MWh
In-orbit assembly and deployment cost	£70.8/MWh	£15.2/MWh
Space-generation hardware cost	£72.5/MWh	£13.5/MWh
Initial programme contingency	£73.2/MWh	£12.8/MWh
Construction duration	£74.7/MWh	£11.4/MWh
Staging-to-operational-orbit transfer cost	£78.4/MWh	£7.6/MWh

£80: Delivered capacity factor=0.9705, £80: Operating lifetime after commissioning=43, £80: Real project discount rate=0.05853, £60: Real project discount rate=0.03423, £80: Construction duration=1, £80: System specific mass=4.504, £60: System specific mass=2.861, £80: Space-generation hardware cost=0.244

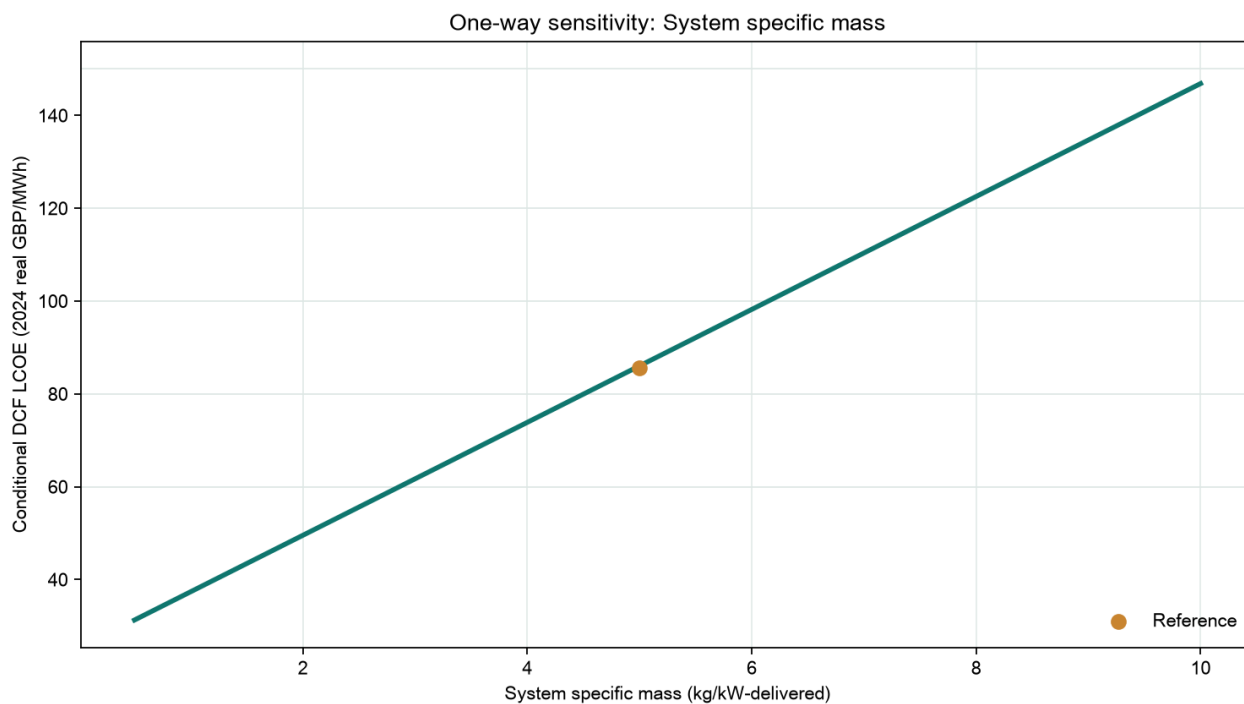
The equal-fraction frontier is a mathematical interpolation device, not a readiness score, probability, forecast, schedule or engineering roadmap.

Target	Equal-fraction movement	Computed chain efficiency
£150/MWh	0.0%	14.98%
£120/MWh	0.0%	14.98%
£100/MWh	0.0%	14.98%
£80/MWh	2.9%	15.36%
£60/MWh	12.5%	16.64%

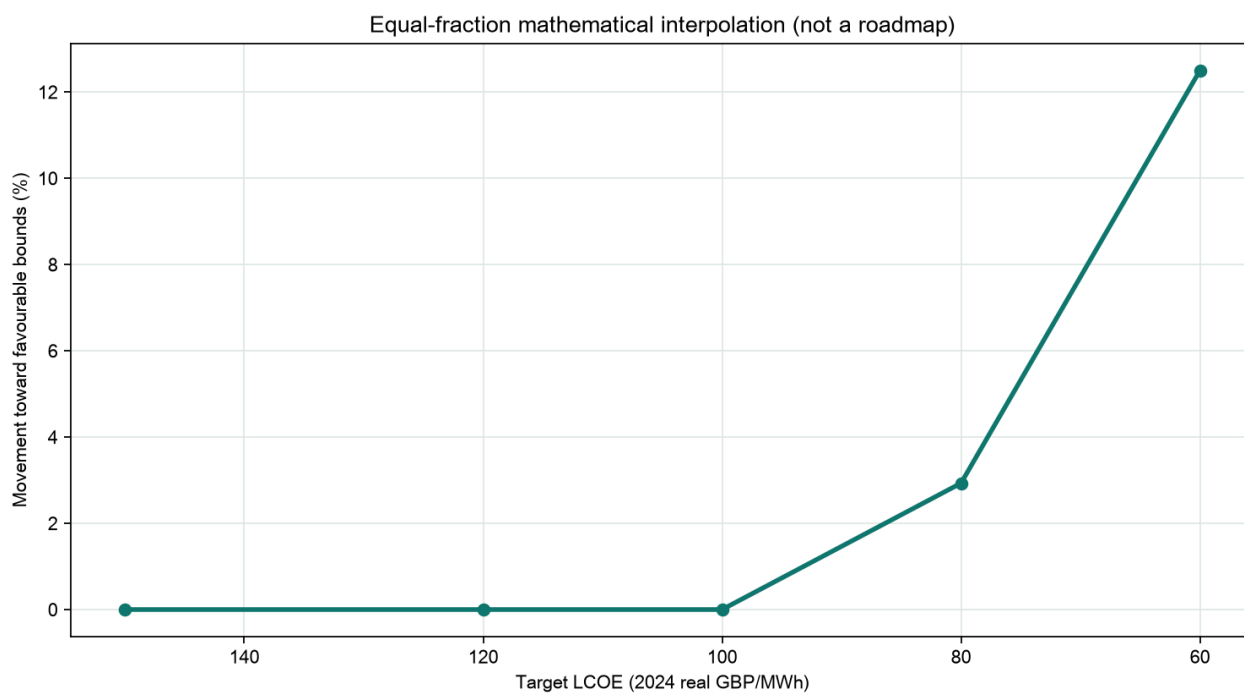
One-way floors



Mass sensitivity



Combined interpolation



Parameter evidence and limitations

Every user-facing numerical input below carries source type, source ID, price year where relevant, denominator and limitation metadata.

Parameter	Reference	Range	Source	Denominator / boundary	Limitation
Delivered grid capacity	2000 MW-delivered AC	500–5000	exploratory / ASSUMPTION_THIS_STUDY	AC power at the grid connection point	Scale input; the simplified model is nearly scale-neutral.
Solar-to-DC conversion efficiency	0.35 fraction	0.2–0.45	exploratory / ASSUMPTION_THIS_STUDY	Incident solar power to space DC-bus output	Architecture-dependent exploratory value; not an independently verified flight-system value.
DC-to-RF efficiency	0.7 fraction	0.4–0.85	exploratory / ASSUMPTION_THIS_STUDY	Space DC-bus input to emitted RF power	Excludes propagation and rectenna conversion losses.
RF transmission efficiency	0.98 fraction	0.85–0.995	exploratory / ASSUMPTION_THIS_STUDY	Emitted RF power to RF incident on the rectenna	Simplified beam-capture proxy; beam geometry sidelobes weather and safety constraints are not explicitly modelled.
Rectenna RF-to-DC efficiency	0.65 fraction	0.4–0.85	exploratory / ASSUMPTION_THIS_STUDY	Incident RF power at rectenna to rectenna DC output	Exploratory system-average conversion proxy.
DC-to-grid AC efficiency	0.96 fraction	0.9–0.99	exploratory / ASSUMPTION_THIS_STUDY	Rectenna DC output to AC at the grid connection point	Includes simplified inversion and local electrical losses only.
Delivered capacity factor	0.9 fraction	0.6–0.98	exploratory / ASSUMPTION_THIS_STUDY	Annual AC energy divided by rated delivered AC capacity times 8760	Combines availability outages and operating constraints; not a measured forecast.
Operating lifetime after commissioning	30 years	15–45	exploratory / ASSUMPTION_THIS_STUDY	Full operating years after commissioning	Engineering retirement and economic life are simplified into one duration.
Real project discount rate	0.065 fraction	0.03–0.2	exploratory / ASSUMPTION_THIS_STUDY	Discounting from start of construction $t=0$	Not an observed SBSP WACC; tax financing tranches and inflation are excluded.
Construction duration	4 years	0–8	exploratory / ASSUMPTION_THIS_STUDY	Years from valuation base to commissioning	Default uses equal annual spend shares; project-specific scheduling is not represented.
Annual delivered-output degradation	0.005 fraction/year	0–0.03	exploratory / ASSUMPTION_THIS_STUDY	Year-on-year reduction in delivered AC energy after the first operating year	Simplified fleet-average degradation proxy.
System specific mass	5 kg/kW-delivered	0.5–10	exploratory / ASSUMPTION_THIS_STUDY	Complete operational orbital hardware mass per kW delivered AC at the grid boundary	Delivered-power basis; never divide by efficiency again. Published values can be architecture-specific or manufacturer-claimed.

Parameter	Reference	Range	Source	Denominator / boundary	Limitation
Space-generation hardware cost	0.4 GBP/W-DC	0.05–5	exploratory / ASSUMPTION_THIS_STUDY	Space DC-bus rated output excluding RF transmitter hardware	Includes collection structure conversion and DC conditioning only; boundary is mutually exclusive with transmitter cost.
RF transmitter hardware cost	0.1 GBP/W-RF emitted	0.02–1	exploratory / ASSUMPTION_THIS_STUDY	Emitted RF rated power excluding generation and DC-bus hardware	Includes only DC-to-RF and aperture hardware; no generation hardware is counted here.
Launch service cost to staging orbit	500 GBP/kg to staging orbit	20–5000	exploratory / ASSUMPTION_THIS_STUDY	Operational orbital hardware mass delivered to the defined LEO staging boundary	Not a GEO delivery quotation; procurement cadence integration and destination materially affect price.
Launch price per flight	1e+08 GBP/flight	1e+07–3e+08	exploratory / ASSUMPTION_THIS_STUDY	One mutually exclusive launch flight to the staging orbit	Used only when per-flight mode is active; never added to per-kg launch cost.
Effective payload per flight	100000 kg/flight	20000–250000	exploratory / ASSUMPTION_THIS_STUDY	Maximum manifested hardware mass reaching the staging orbit per flight	Vehicle and orbit dependent; used for launch-count diagnostics in both modes.
Payload utilisation	0.85 fraction	0.5–1	exploratory / ASSUMPTION_THIS_STUDY	Average used fraction of effective payload capability	Does not model packaging volume schedule or rideshare constraints.
Staging-to-operational-orbit transfer cost	100 GBP/kg final hardware	0–3000	exploratory / ASSUMPTION_THIS_STUDY	Final operational hardware moved from staging orbit to operational orbit	Must cover transfer vehicle propellant refuelling missions operations and payload penalty; a low value does not prove cheap HEO delivery.
In-orbit assembly and deployment cost	200 GBP/kg operational hardware	0–3000	exploratory / ASSUMPTION_THIS_STUDY	Operational orbital hardware assembled deployed and commissioned	Exploratory allowance; robotic operations inspection spares and commissioning are bundled.
Rectenna cost	0.15 GBP/W-delivered AC	0.02–1	exploratory / ASSUMPTION_THIS_STUDY	Rated AC power delivered at the grid boundary	Bundled proxy; antenna area frequency beam geometry sidelobes power-density land weather and safety are not explicit.
Grid connection cost	100 GBP/kW-delivered AC	20–300	exploratory / ASSUMPTION_THIS_STUDY	Rated AC power at the grid connection point	Simplified local connection allowance; wider transmission reinforcement is excluded.
Initial programme contingency	0.2 fraction of pre-contingency initial CAPEX	0–0.5	exploratory / ASSUMPTION_THIS_STUDY	Applied once to pre-contingency initial construction CAPEX	Not applied recursively to annual O&M or replacement allowances.

Parameter	Reference	Range	Source	Denominator / boundary	Limitation
Space-hardware replacement rate	0.006 fraction/year	0–0.03	exploratory / ASSUMPTION_THIS_STUDY	Fraction of eligible space hardware plus associated launch transfer and assembly cost replaced each year	Simplified annual equivalent rather than discrete failure and logistics scheduling.
Ground-hardware replacement rate	0.003 fraction/year	0–0.02	exploratory / ASSUMPTION_THIS_STUDY	Fraction of eligible rectenna and grid cost replaced each year	Does not model component-specific replacement intervals.
Fixed O&M rate	0.01 fraction/year	0.002–0.05	exploratory / ASSUMPTION_THIS_STUDY	Applied to generation transmitter rectenna and grid assets only	Excludes contingency initial launch transfer and assembly from its base.
Variable O&M	3 GBP/MWh delivered	0–20	exploratory / ASSUMPTION_THIS_STUDY	Each MWh of AC energy delivered at the grid boundary	Exploratory allowance; market charges and downstream system costs are excluded.
Terminal decommissioning cost	0.02 fraction of initial CAPEX	0–0.2	exploratory / ASSUMPTION_THIS_STUDY	Applied at the end of operating life to initial CAPEX	Timing and disposal obligations are simplified.
Terminal residual value	0 fraction of initial CAPEX	0–0.2	exploratory / ASSUMPTION_THIS_STUDY	Credit at end of operating life relative to initial CAPEX	No default credit; salvage markets and liabilities are uncertain.

Formulae, boundaries and validation

Mass is orbital mass = delivered capacity (kW) × kg/kW-delivered. Hardware costs are rated independently at space-DC-bus W, emitted-RF W, delivered W or delivered kW, with mutually exclusive boundaries. Space replacement includes hardware plus associated launch, transfer and assembly; ground replacement includes only rectenna and grid. Fixed O&M excludes contingency, initial launch, transfer and assembly.

The default construction profile is [0.25, 0.25, 0.25, 0.25] and sums to 100%. Programme contingency is applied once to initial CAPEX. Terminal decommissioning is a cost and residual value is a credit at the end of operating life.

The test suite checks reciprocal mass conversion, the 2 GW / 0.67 kW/kg regression, no second efficiency division, all six stage powers, rated-cost assignment, launch-mode exclusivity, launch rounding, replacement bases, construction shares, DCF/CRF convergence, zero-rate handling, invalid inputs, Python/browser parity, bilingual parity, removal of the historical mass identifier and full regeneration.

Remaining limitations

- No physical design of antenna area, frequency, beam geometry, sidelobes, power density, land, weather or safety constraints.
- No vehicle manifest, structural load, thermal, radiation, failure, spares, discrete replacement mission or detailed launch schedule.
- No tax, inflation, financing tranches, learning curves or architecture-parameter correlations.
- No GB dispatch, network reinforcement, balancing, capacity value or market-revenue model.
- Exploration bounds are not probability distributions; no P10/P50/P90 labels are produced.

Generated from the executable model. Price year: 2024 real GBP. Valuation base: start of construction (t=0).